

Resolution Geometry of Observable Transport Systems

A Geometric Framework for Independent Reconstruction of Transport Invariants

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Abstract

Many observational sciences infer transport structure from local response measurements. Typical analyses construct a single geometric object, such as a metric, connection, covariance model, or information geometry, and study its properties independently.

This work develops an alternative viewpoint. Instead of treating these structures as primary, we introduce the *resolution geometry* associated with observable transport. The construction requires only a differentiable local response operator and an empirical observation covariance. From these data the observable space decomposes canonically into resolved and unresolved sectors, producing a minimal geometric object

$$\mathcal{G} = (O; R, N, \Sigma).$$

Admissible transformations are defined intrinsically as those preserving this geometry. The resulting morphisms form a category of observable resolution geometries, from which several familiar computational structures arise naturally as derived invariant functors, including information metrics, covariance decompositions, null-space geometry, and transport holonomy whenever additional transport structure exists.

A computational protocol is developed in which multiple invariant families are reconstructed independently from identical observational data. Rather than attempting to validate a physical model, the framework tests whether these independently reconstructed invariants recover mutually compatible descriptions of the same observable transport geometry.

The methodology is illustrated on reproducible synthetic transport systems, where multiple computational batteries evaluate different invariant families from a common dataset. Agreement between independently reconstructed invariants demonstrates the internal coherence of the proposed geometric framework, while deliberately constructed counterexamples illustrate its sensitivity to violations of the underlying resolution geometry.

The framework is formulated independently of any particular physical interpretation and is intended as an observational geometry for transport systems. Questions concerning globalization, covariance coupling, and physical realizations remain open mathematical problems.

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1 Introduction

Observable transport systems appear throughout mathematics, physics, engineering, and the natural sciences. Although the governing mechanisms differ substantially between applications, they share a common operational feature: small perturbations generate measurable responses from which local transport structure may be inferred.

Existing geometric descriptions typically begin by selecting one mathematical structure as fundamental. Differential geometry begins with manifolds, metrics, and connections. Information geometry begins with statistical models and monotone metrics. Inverse problems often emphasize sensitivity operators or Jacobians. Covariance-based methods focus on uncertainty propagation. Transport theory introduces flows or evolution operators.

These approaches have proved highly successful within their respective domains. However, they generally investigate one geometric representation at a time. The relationship between independently reconstructed structures obtained from identical observational data remains comparatively unexplored.

The present work adopts a different perspective.

Rather than postulating a metric, connection, or transport law, we ask a more elementary question:

What is the minimal observable geometry determined solely by local response measurements and empirical observational uncertainty?

The answer proposed here is the *resolution geometry*

$$\mathcal{G} = (O; R, N, \Sigma),$$

consisting of the observable space, the resolved sector induced by the local response operator, its complementary unresolved sector, and the empirical observation covariance.

This object is operational rather than ontological. It depends only upon measurable quantities and requires no assumptions concerning hidden variables or microscopic mechanisms.

Once the resolution geometry is established, admissible transformations are defined intrinsically as those preserving its structure. This permits the development of a category of observable geometries together with a hierarchy of derived invariant functors.

The central computational hypothesis investigated throughout this paper is not that a particular physical theory is correct. Instead, we examine whether multiple independently reconstructed invariant families recover mutually compatible descriptions of the same observable geometry when computed from identical observational data.

To investigate this question, we develop a reproducible computational protocol based on synthetic transport systems. Every computational battery receives exactly the same observational dataset, allowing agreement or disagreement between invariant families to be attributed solely to their mathematical construction.

The principal contributions of the paper are as follows.

1. Definition of the resolution geometry as a minimal observable geometric object.

2. Characterization of admissible morphisms preserving this geometry.
3. Construction of invariant functors associated with observable transport.
4. Development of an independent computational protocol for reconstructing multiple invariant families from identical observational data.
5. Systematic comparison of these invariant families on reproducible synthetic transport systems.
6. Identification of several open mathematical problems concerning globalization, covariance coupling, and transport holonomy.

The framework developed here is referred to as *Compatible Transport Morphing Theory* (CTMT). Within the present paper, however, CTMT is interpreted primarily as an observational geometric framework rather than as a physical theory. Its objective is to organize measurable transport structures into a coherent geometric language while remaining agnostic concerning their ultimate physical realization.

1.1 Motivation

Several independent investigations motivated the present framework.

When identical synthetic transport systems were analysed through different computational batteries, remarkably different notions of closure emerged.

Among them were

- unit closure,
- Jacobian closure,
- Fisher-information closure,
- null-space closure,
- covariance closure,
- transport holonomy,
- winding invariants.

Although these quantities were computed independently, they repeatedly exhibited strong structural correlations.

This observation suggested that the individual diagnostics were not fundamental objects, but rather projections of a common underlying geometry.

The principal objective of the present work is therefore not the construction of new invariants.

Instead, the goal is to identify the minimal mathematical structure from which these operational invariants arise naturally.

1.2 Scope

The present paper develops only the mathematical foundations.

No claim is made that the proposed geometry constitutes a universal physical description.

Instead, the following more modest questions are addressed.

1. What geometric information can be reconstructed from observable response alone?

2. How does empirical covariance modify this reconstruction?
3. How are resolved and unresolved directions organized?
4. Which transformations preserve the resulting geometry?
5. How do multiple operational notions of closure arise from a common construction?

Questions concerning physical interpretation, globalization, and covariance-induced transport coupling remain open and are discussed separately.

1.3 Relationship to Existing Geometry

The present framework does not replace differential geometry, information geometry or transport theory.

Instead, it occupies an intermediate operational level.

Differential geometry studies structures defined on manifolds.

Information geometry studies statistical manifolds equipped with monotone metrics.

Transport theory studies evolution under prescribed constitutive laws.

Here the starting point is different.

Observable transport induces local response operators. Response operators determine resolved directions. Measurement covariance determines observational resolution. Together these define an operational geometry before any particular metric, connection or constitutive model is imposed.

Whenever additional physical structure exists, it may induce special cases of the general construction developed below.

2 Observable Transport Systems

We now introduce the primitive mathematical objects.

Throughout the paper no assumption is made regarding the physical origin of the observations.

The framework applies equally to synthetic systems, laboratory measurements, inverse problems, and numerical simulations, provided local perturbations of observables can be estimated.

Definition 2.1 (Observation space). An observation space is a finite-dimensional real vector space

$$O \cong \mathbb{R}^m$$

whose coordinates represent experimentally accessible observables.

No assumption is made regarding the physical interpretation of these coordinates.

An observable state is therefore represented only through its measurable coordinates.

Internal variables inaccessible to observation play no role in the present construction.

Definition 2.2 (Local response operator). Let

$$x \in O$$

be an observable state.

The local response operator at x is the Jacobian

$$J(x) = \frac{\partial y}{\partial x},$$

computed from admissible perturbations of observable quantities.
 Its image represents locally distinguishable first-order response directions.
 Its null space represents directions unresolved to first order.

The local response operator constitutes the primary source of geometric information throughout the framework.

Definition 2.3 (Empirical covariance). Let

$$y = (y_1, \dots, y_m)^T$$

denote the observable response vector associated with a local experiment.
 The empirical covariance operator is the symmetric positive semi-definite matrix

$$\Sigma = \text{Cov}(y) \in \mathbb{R}^{m \times m},$$

estimated from repeated realizations of the same operational protocol.

Throughout this work, covariance is regarded as an operational property of the observation process rather than an intrinsic property of the underlying physical system.

Unlike the local response operator, which describes first-order distinguishability, the covariance operator characterizes uncertainty in the observation process.

Both objects are required for the subsequent geometric construction.

2.1 Resolved and Unresolved Directions

The response operator naturally induces a decomposition of observation space.

Only part of the observable space contributes to first-order distinguishability. The remaining directions are locally invisible under infinitesimal perturbation.

This distinction forms the basis of the operational geometry introduced in the following sections.

Definition 2.4 (Resolved sector). The resolved sector is the image of the local response operator,

$$R = \text{Im}(J).$$

Elements of R correspond to first-order observable directions generated by admissible perturbations.

Definition 2.5 (Null sector). The null sector is defined by

$$N = \text{Ker}(J^T).$$

Equivalently,

$$N = R^\perp$$

with respect to the Euclidean inner product on the observation space.

Elements of N correspond to observable directions that remain indistinguishable to first order.

The choice of the left null space rather than the kernel of J is deliberate.

The construction concerns observables rather than parameter directions. Consequently, the natural decomposition occurs inside the observation space itself.

Theorem 2.6 (Resolved–Null Decomposition). *Let*

$$J : \mathbb{R}^n \rightarrow \mathbb{R}^m$$

*be a linear response operator.
Then*

$$R = \text{Im}(J)$$

and

$$N = \text{Ker}(J^T)$$

form complementary orthogonal subspaces satisfying

$$O = R \oplus N.$$

Furthermore,

$$R \cap N = \{0\},$$

and

$$\dim(R) + \dim(N) = m.$$

Proof. The result follows directly from the Fundamental Theorem of Linear Algebra.
Indeed,

$$\text{Ker}(J^T) = \text{Im}(J)^\perp,$$

and every finite-dimensional Euclidean vector space decomposes uniquely into a subspace and its orthogonal complement.

Therefore,

$$O = \text{Im}(J) \oplus \text{Ker}(J^T).$$

The remaining statements follow immediately from orthogonality.

□

The preceding theorem provides the first structural result of the framework.

No statistical assumptions have been introduced.

No metric beyond the Euclidean structure of observation space has been required.

The decomposition therefore exists for every observable transport system possessing a differentiable local response operator.

2.2 Covariance Decomposition

Once the resolved and null sectors have been identified, the covariance operator inherits a natural block structure.

Choosing an orthonormal basis adapted to the decomposition

$$O = R \oplus N,$$

the covariance matrix assumes the form

$$\Sigma = \begin{pmatrix} \Sigma_R & C_{RN} \\ C_{RN}^T & \Sigma_N \end{pmatrix}.$$

Here

- Σ_R denotes covariance restricted to the resolved sector,
- Σ_N denotes covariance restricted to the null sector,
- C_{RN} denotes resolved–null covariance coupling.

This decomposition is canonical once the response operator has been fixed.

Definition 2.7 (Resolved–Null Coupling). The matrix

$$C_{RN}$$

will be called the resolved–null covariance coupling.
When

$$C_{RN} = 0,$$

resolved and unresolved uncertainties are statistically decoupled.
When

$$C_{RN} \neq 0,$$

the observational covariance exhibits cross-sector interaction.

No physical interpretation of this coupling is assumed in the present work.

Its operational significance is investigated through automorphism classification in later sections.

Remark 2.8. The covariance decomposition should not be interpreted as a decomposition of the underlying physical system.

Rather, it is a decomposition of observational uncertainty relative to the resolved and unresolved directions determined by the local response operator.

Consequently, two physically distinct systems may induce identical resolution geometries, while identical physical systems observed under different protocols may induce different covariance decompositions.

This distinction is fundamental throughout the paper.

3 Resolution Geometry

The preceding constructions motivate the introduction of the central mathematical object.

Unlike classical differential geometry, where the metric tensor is postulated, the present framework derives its primitive object from the pair consisting of local response and empirical covariance.

Definition 3.1 (Resolution Geometry). The local resolution geometry associated with an observable transport system is the quadruple

$$\boxed{\mathcal{G} = (O; R, N, \Sigma)}.$$

where

$$R = \text{Im}(J), \quad N = \text{Ker}(J^T),$$

and

$$\Sigma$$

is the empirical covariance operator expressed with respect to the induced resolved–null decomposition.

Resolution geometry is regarded as the primitive mathematical object of the CTMT framework.

Subsequent notions of admissible transport, closure, information geometry, null-sector analysis, and transport holonomy will all be formulated as properties derived from $\mathcal{G}$, rather than introduced independently.

4 Admissible Morphisms

The resolution geometry introduced in the previous section provides the primitive object of the framework.

The next natural question is therefore not how to compute additional invariants, but rather which transformations preserve the induced geometry.

The present section introduces this notion.

4.1 Pullback of Resolution Geometry

Let

$$\phi : O_1 \rightarrow O_2$$

be a smooth invertible transformation between observation spaces.

The differential

$$D\phi$$

induces the usual pullback of vector subspaces and tensor fields.

Consequently,

the image,

null sector,

and covariance transform naturally.

The pullback of the covariance operator is

$$\phi^* \Sigma = (D\phi)^T \Sigma (D\phi),$$

while the resolved and null sectors satisfy

$$\phi^* R = (D\phi)^{-1}(R),$$

and

$$\phi^* N = (D\phi)^{-1}(N).$$

These constructions are entirely standard.

The novelty lies not in the pullback itself, but in identifying the complete quadruple

$$(O; R, N, \Sigma)$$

as the primitive object whose preservation defines admissible transport.

Definition 4.1 (Admissible Morphism). Let

$$\mathcal{G}_1 = (O_1; R_1, N_1, \Sigma_1)$$

and

$$\mathcal{G}_2 = (O_2; R_2, N_2, \Sigma_2)$$

be two local resolution geometries.

A smooth invertible map

$$\phi : O_1 \rightarrow O_2$$

is called an admissible morphism whenever

$$\boxed{\phi^* \mathcal{G}_2 = \mathcal{G}_1.}$$

Explicitly,
this requires

$$\phi^* R_2 = R_1,$$

$$\phi^* N_2 = N_1,$$

and

$$\phi^* \Sigma_2 = \Sigma_1.$$

Unlike optimization-based approaches, no variational principle is assumed.

Admissibility is determined entirely by preservation of the observable resolution geometry.

Remark 4.2. The definition above should be regarded as the analogue of metric preservation in Riemannian geometry or symplectic-form preservation in symplectic geometry.

The present framework does not begin with a metric tensor.

Instead, admissible morphisms preserve the operational decomposition induced jointly by local response and observational covariance.

4.2 Automorphism Group

The preceding definition immediately induces a natural symmetry group.

Definition 4.3. The automorphism group of a local resolution geometry is

$$\boxed{\text{Aut}(\mathcal{G}) = \{\phi \in \text{Diff}(O) \mid \phi^* \mathcal{G} = \mathcal{G}\}.}$$

The automorphism group characterizes all transformations preserving the observable geometry.

It therefore replaces the role traditionally played by isometry groups in metric geometry.

Proposition 4.4. *Every automorphism preserves*

1. *the dimension of the resolved sector,*
2. *the dimension of the null sector,*
3. *the rank of the response operator,*
4. *the covariance block decomposition.*

Proof. Since admissible morphisms preserve

$$R = \text{Im}(J),$$

they preserve its dimension.

Likewise,

preservation of

$$N = \text{Ker}(J^T)$$

preserves the nullity.

The rank-nullity theorem then immediately implies preservation of the rank.

Finally,

preservation of the covariance operator together with preservation of the sector decomposition preserves the block representation

$$\Sigma = \begin{pmatrix} \Sigma_R & C_{RN} \\ C_{RN}^T & \Sigma_N \end{pmatrix}.$$

□

Corollary 4.5. *Every scalar quantity computed solely from*

$$(R, N, \Sigma)$$

is invariant under admissible morphisms.

The corollary explains why seemingly unrelated operational diagnostics may agree.

If they depend only upon the preserved resolution geometry, they necessarily belong to the same automorphism class.

4.3 Rigidity Induced by Sector Coupling

The covariance decomposition reveals an important distinction between decoupled and coupled observable geometries.

When

$$C_{RN} = 0,$$

resolved and null sectors become statistically independent.

When

$$C_{RN} \neq 0,$$

cross-sector covariance constrains admissible transformations.

This observation motivates the following theorem.

Theorem 4.6 (Automorphism Splitting). *Suppose*

$$C_{RN} = 0.$$

Then every automorphism acts independently on the resolved and null sectors. Consequently,

$$\text{Aut}(\mathcal{G}) \supseteq \text{Aut}(R, \Sigma_R) \times \text{Aut}(N, \Sigma_N).$$

Proof. If

$$C_{RN} = 0,$$

the covariance operator is block diagonal with respect to

$$O = R \oplus N.$$

Since admissible morphisms preserve both the decomposition and the covariance, their action on each invariant subspace is independent.

Therefore the automorphism group contains the direct product of the automorphism groups associated with each block. □

Remark 4.7. Whether the above inclusion is an equality depends upon additional global structure and is left as an open problem.

Likewise,
the physical interpretation of non-vanishing coupling

$$C_{RN}$$

remains unresolved.

The present work establishes only its geometric role within the local automorphism classification.

5 Derived Structures of Resolution Geometry

The preceding sections establish the local resolution geometry

$$\mathcal{G} = (O; R, N, \Sigma)$$

as the primitive object of the framework.

No metric, connection, or transport invariant has yet been introduced.

The purpose of the present section is to demonstrate that several familiar objects arise naturally as structures induced by $\mathcal{G}$.

Rather than defining multiple independent geometries, the framework regards them as different projections of the same observable resolution geometry.

5.1 Response Metric

The local response operator

$$J : T_x M \rightarrow O$$

induces the positive semi-definite quadratic form

$$F = J^T \Sigma^{-1} J,$$

whenever the covariance operator is invertible on the resolved sector.

This matrix coincides with the classical Fisher information matrix in statistical settings.

Within the present framework, however, its interpretation is purely geometric.

It measures the distinguishability of infinitesimal perturbations after projection through the observable response operator.

Definition 5.1 (Response Metric). The response metric associated with a local resolution geometry is

$$g_R = J^T \Sigma^{-1} J.$$

The response metric depends only upon the observable response operator and the empirical covariance.

No probabilistic model is assumed.

Remark 5.2. When the observations arise from a regular statistical model satisfying the classical assumptions of information geometry, the response metric coincides with the Fisher information metric.

Consequently, information geometry appears as a special realization of the more general resolution geometry.

5.2 Null Geometry

The null sector

$$N = \text{Ker}(J^T)$$

contains observable directions that remain invisible to first-order perturbation.

Although these directions do not contribute to the response metric, they generally possess non-trivial covariance.

This motivates a second induced structure.

Definition 5.3 (Null Covariance). The null covariance is the restriction

$$\boxed{\Sigma_N = P_N \Sigma P_N},$$

where

$$P_N$$

denotes the orthogonal projector onto the null sector.

Unlike the response metric, which measures distinguishability, the null covariance characterizes unresolved observational variability.

Remark 5.4. The null covariance should not be interpreted as noise.

It represents uncertainty remaining after projection onto directions that are locally invisible to first order.

Consequently, large null covariance does not imply poor measurements.

Rather, it indicates unresolved structure relative to the current observable response operator.

5.3 Resolved–Null Coupling

The covariance decomposition

$$\Sigma = \begin{pmatrix} \Sigma_R & C_{RN} \\ C_{RN}^T & \Sigma_N \end{pmatrix}$$

contains the coupling block

$$C_{RN},$$

which links resolved and unresolved observational directions.

The following proposition summarizes its geometric role.

Proposition 5.5. *The resolved–null coupling vanishes if and only if the covariance operator preserves the orthogonal decomposition*

$$O = R \oplus N.$$

Proof. If

$$C_{RN} = 0,$$

the covariance matrix is block diagonal.
Hence both

$$R$$

and

$$N$$

are invariant under the covariance operator.

Conversely, any covariance operator preserving both invariant subspaces possesses a block diagonal representation, forcing

$$C_{RN} = 0.$$

□

5.4 Operational Closure

Several closure notions introduced later are naturally expressed using the resolution geometry.

Rather than defining each independently, we regard closure as preservation of a specified component of $\mathcal{G}$.

This viewpoint leads to a hierarchy.

Definition 5.6 (Closure Hierarchy). Let

$$\phi$$

be an admissible transformation.

Different notions of operational closure correspond to preservation of different derived structures.

Examples include

1. response closure;
2. covariance closure;
3. null-sector closure;
4. response-metric closure;
5. transport closure.

Each notion is therefore interpreted as preservation of a particular projection of the same underlying resolution geometry.

The principal philosophy of the framework may now be summarized as follows.

Observable transport induces a response operator.
The response operator induces a decomposition of observation space.
The decomposition together with empirical covariance defines a local resolution geometry.
Metrics, null structures, transport invariants, and operational closure are derived from this geometry rather than introduced independently.

6 Canonical Properties of Resolution Geometry

The construction developed in the preceding sections is intentionally minimal.

Only three operational objects are assumed:

1. an observation space,
2. a differentiable local response operator,
3. an empirical covariance operator.

No metric, connection, transport equation, constitutive relation, or variational principle has been imposed.

The purpose of the present section is to establish that the resulting resolution geometry is a canonical consequence of these assumptions rather than an additional postulate.

6.1 Coordinate Independence

A natural concern is whether the decomposition

$$O = R \oplus N$$

depends upon the particular coordinate representation of the observations.

The following proposition shows that it does not.

Proposition 6.1. *Let*

$$\psi : O \rightarrow O$$

be an invertible smooth change of observable coordinates.

Then

$$\psi(R) = R',$$

and

$$\psi(N) = N',$$

where

$$R' = \text{Im}(J'), \quad N' = \text{Ker}(J'^T),$$

are computed from the transformed response operator.

Consequently, the decomposition of observation space is coordinate invariant.

Proof. Under a change of coordinates,

$$J' = D\psi \circ J.$$

Since

$$D\psi$$

is invertible,

$$\text{Im}(J') = D\psi(\text{Im}(J)).$$

Likewise,

$$J'^T = J^T (D\psi)^T,$$

implying

$$\text{Ker}(J'^T) = D\psi(\text{Ker}(J)).$$

Therefore both subspaces transform covariantly under coordinate changes. □

Remark 6.2. The decomposition itself is invariant.

Only its coordinate representation changes.

Accordingly, resolution geometry is attached to the observable system, not to a particular choice of coordinates.

6.2 Minimality

The resolution geometry contains exactly those structures required to describe first-order observable transport.

Removing any component destroys information essential for subsequent construction.

Proposition 6.3. *Each component of*

$$\mathcal{G} = (O; R, N, \Sigma)$$

is independent in the following sense.

1. *Without R , observable distinguishability cannot be recovered.*
2. *Without N , locally unresolved directions cannot be identified.*
3. *Without Σ , measurement resolution is undefined.*
4. *Without O , the remaining objects possess no common ambient space.*

Proof. Each statement follows immediately from the definitions.

None of the four components can be reconstructed uniquely from the remaining three.

Consequently, the quadruple forms a minimal complete description of first-order observable resolution. □

6.3 Functoriality

Resolution geometry behaves naturally under admissible morphisms.

Rather than introducing additional compatibility conditions, the entire construction is inherited from standard pullback operations.

Theorem 6.4. *The assignment*

$$O \mapsto \mathcal{G}(O)$$

together with admissible morphisms

$$\phi : \mathcal{G}_1 \rightarrow \mathcal{G}_2$$

defines a category.

Composition is ordinary composition of smooth maps.

Identity morphisms are admissible.

Composition of admissible morphisms remains admissible.

Proof. Identity maps preserve

$$R, \quad N, \quad \Sigma$$

trivially.
Suppose

$$\phi : \mathcal{G}_1 \rightarrow \mathcal{G}_2$$

and

$$\psi : \mathcal{G}_2 \rightarrow \mathcal{G}_3$$

are admissible.
Then

$$\phi^* \mathcal{G}_2 = \mathcal{G}_1,$$

and

$$\psi^* \mathcal{G}_3 = \mathcal{G}_2.$$

Therefore

$$(\psi \circ \phi)^* = \phi^* \psi^*,$$

implying

$$(\psi \circ \phi)^* \mathcal{G}_3 = \mathcal{G}_1.$$

Hence composition is admissible.

Associativity follows from ordinary composition of smooth maps.

□

Corollary 6.5. *Resolution geometry together with admissible morphisms forms the category*

$$\text{ResGeom}.$$

6.4 Relation to Classical Geometries

The present framework should not be regarded as competing with established geometries.

Instead, several familiar constructions appear naturally as projections or special cases.

1. If the covariance operator is ignored,
the framework reduces to the geometry of linear response.
2. If the response operator arises from a regular statistical model,
the induced information metric coincides with the Fisher information metric.
3. If transport admits a compatible connection,
parallel transport and holonomy become additional derived structures.
4. If a smooth Lorentzian metric exists independently,
resolution geometry becomes an observable structure defined upon that background.

Consequently, the framework neither replaces nor contradicts existing mathematical structures.

Instead, it identifies a common operational geometry from which several familiar objects emerge.

6.5 Fundamental Principle

The previous sections may be summarized by a single defining principle.

Fundamental Principle of Resolution Geometry

Observable transport determines a local response operator.

The response operator induces resolved and unresolved sectors.

Together with empirical covariance, these define the local resolution geometry

$$\mathcal{G} = (O; R, N, \Sigma).$$

Admissible transformations are precisely those preserving this geometry,

$$\phi \in \text{Mor}_{\text{CTMT}} \iff \phi^* \mathcal{G} = \mathcal{G}.$$

All subsequent transport invariants, closure notions, and computational diagnostics are regarded as derived structures of $\mathcal{G}$.

7 Closure Invariants

The resolution geometry introduced in the preceding sections provides a single primitive geometric object.

Different computational procedures generally preserve different subsets of this geometry.

Rather than viewing these procedures as unrelated diagnostics, we interpret them as defining distinct classes of geometric invariants.

The purpose of the present section is to formalize this hierarchy.

7.1 Invariant Functionals

Let

$$\mathcal{G} = (O; R, N, \Sigma)$$

be a local resolution geometry.

Definition 7.1. A functional

$$I(\mathcal{G})$$

is called an invariant of the resolution geometry if

$$I(\phi^* \mathcal{G}) = I(\mathcal{G})$$

for every admissible morphism

$$\phi \in \text{Aut}(\mathcal{G}).$$

Thus every invariant depends only upon the automorphism class of the resolution geometry rather than upon a particular coordinate representation.

Remark 7.2. This definition is entirely analogous to the classical notion of invariant objects in differential geometry.

The distinction is that the primitive object is not a metric or connection, but the observable resolution geometry.

7.2 Examples of Invariants

Several quantities appearing throughout the paper are immediate examples.

Proposition 7.3. *The following quantities are invariants of the local resolution geometry.*

1. *The rank*

$$\text{rank}(J).$$

2. *The dimensions*

$$\dim(R), \quad \dim(N).$$

3. *The eigenvalue spectra of*

$$\Sigma_R, \quad \Sigma_N.$$

4. *The singular values of the response operator.*

5. *The eigenvalue spectrum of the induced information metric*

$$J^T \Sigma^{-1} J.$$

Proof. Each quantity is preserved under admissible pullback because admissible morphisms preserve

$$R, \quad N, \quad \Sigma,$$

and therefore preserve every algebraic quantity computed from these objects. □

7.3 Closure Levels

The previous proposition motivates a hierarchy.

Different computational procedures verify preservation of different collections of invariants.

Definition 7.4. A closure level is a collection

$$\mathcal{C} = \{I_1, I_2, \dots, I_k\}$$

of invariants required to remain unchanged under admissible transport.

Different choices of

$$\mathcal{C}$$

lead naturally to different notions of closure.

The hierarchy investigated in this work consists of

1. Rank closure,
2. Resolved-sector closure,
3. Null-sector closure,

4. Covariance closure,
5. Information-metric closure,
6. Holonomy closure.

Each successive level preserves strictly more of the observable geometry.

Proposition 7.5. *The closure hierarchy is nested.*

If

$$\mathcal{C}_2 \supseteq \mathcal{C}_1,$$

then every transformation satisfying

$$\mathcal{C}_2$$

necessarily satisfies

$$\mathcal{C}_1.$$

Proof. Immediate from the definition.

Preserving a larger family of invariants automatically preserves every subfamily.

□

7.4 Closure Lattice

Closure notions are naturally partially ordered by inclusion of preserved invariants.

Consequently they form a lattice.

$$\text{Rank} \subset \text{Resolved} \subset \text{Covariance} \subset \text{Information} \subset \text{Transport}.$$

This ordering should not be interpreted as measuring physical importance.

It merely quantifies the amount of observable geometry preserved.

Remark 7.6. The existence of several closure notions is therefore not a defect of the framework.

Instead, it reflects the fact that different computational procedures probe different projections of the same geometric object.

The computational batteries developed later in this paper should therefore be interpreted as measurements of different invariant families rather than as independent tests.

7.5 Computational Interpretation

Suppose that an observable transport system is analyzed by several independent numerical procedures.

Examples include

- response linearization,
- information-metric estimation,
- null-space projection,
- covariance decomposition,
- loop transport,
- holonomy estimation.

Although each procedure computes different numerical quantities, all depend ultimately upon the same underlying resolution geometry.

Agreement between these procedures therefore provides evidence that they are sampling a common geometric structure rather than unrelated numerical artifacts.

8 Globalization and Transport Groupoid

The preceding sections develop the resolution geometry $\mathcal{G} = (O; R, N, \Sigma)$ as a local object attached to a single observable state. In many operational settings, however, observations are collected across multiple experimental configurations, spatial locations, or temporal snapshots. This motivates the globalization of the local construction.

The purpose of this section is to introduce charts, overlaps, and the transport groupoid associated with resolution geometry. No global manifold structure is assumed. Instead, globalization proceeds through compatible families of local resolution geometries.

8.1 Charts of Resolution Geometry

Let $\mathcal{U} = \{U_\alpha\}_{\alpha \in A}$ be a collection of observable neighborhoods. Each neighborhood carries its own local resolution geometry

$$\mathcal{G}_\alpha = (O_\alpha; R_\alpha, N_\alpha, \Sigma_\alpha).$$

Definition 8.1 (Resolution Chart). A resolution chart is a pair

$$(U_\alpha, \mathcal{G}_\alpha)$$

consisting of an observable neighborhood and its associated local resolution geometry.

Charts need not arise from coordinates on a manifold. They represent operational domains in which local response and covariance can be estimated reliably.

8.2 Overlaps and Transition Maps

Whenever two observable neighborhoods intersect,

$$U_\alpha \cap U_\beta \neq \emptyset,$$

their resolution geometries must be compared.

Definition 8.2 (Transition Map). A transition map between charts is a smooth invertible transformation

$$\phi_{\alpha\beta} : O_\beta \rightarrow O_\alpha$$

defined on the overlap $U_\alpha \cap U_\beta$, such that

$$\phi_{\alpha\beta}^* \mathcal{G}_\alpha = \mathcal{G}_\beta.$$

Thus transition maps are precisely the admissible morphisms introduced earlier. They preserve resolved directions, null directions, and covariance structure across overlapping charts.

8.3 Cocycle Condition

On triple overlaps,

$$U_\alpha \cap U_\beta \cap U_\gamma \neq \emptyset,$$

transition maps satisfy the usual cocycle condition.

Proposition 8.3 (Cocycle Condition). *For any triple of charts with non-empty triple overlap,*

$$\phi_{\alpha\beta} \circ \phi_{\beta\gamma} = \phi_{\alpha\gamma}.$$

Proof. Each transition map preserves resolution geometry. Therefore,

$$\phi_{\alpha\beta}^* \mathcal{G}_\alpha = \mathcal{G}_\beta, \quad \phi_{\beta\gamma}^* \mathcal{G}_\beta = \mathcal{G}_\gamma.$$

Composition yields

$$(\phi_{\alpha\beta} \circ \phi_{\beta\gamma})^* \mathcal{G}_\alpha = \mathcal{G}_\gamma,$$

which is precisely the defining property of $\phi_{\alpha\gamma}$. Uniqueness of admissible morphisms on overlaps implies equality. $\square$

8.4 Transport Groupoid

The collection of charts and transition maps forms a groupoid.

Definition 8.4 (Resolution Transport Groupoid). The resolution transport groupoid is the category

$$\mathcal{T}(\mathcal{G})$$

whose objects are resolution charts $(U_\alpha, \mathcal{G}_\alpha)$, and whose morphisms are transition maps $\phi_{\alpha\beta}$ defined on overlaps.

- Objects: local resolution geometries.
- Morphisms: admissible transformations preserving resolution geometry.
- Composition: chart transition composition.
- Identity: identity morphisms on each chart.

This groupoid encodes all admissible observable transport between charts.

8.5 Holonomy of Resolution Geometry

Given a sequence of overlapping charts forming a loop,

$$(U_{\alpha_1}, U_{\alpha_2}, \dots, U_{\alpha_k}, U_{\alpha_1}),$$

the composition of transition maps yields an automorphism

$$\Phi = \phi_{\alpha_1\alpha_k} \circ \dots \circ \phi_{\alpha_2\alpha_1} \in \text{Aut}(\mathcal{G}_{\alpha_1}).$$

Definition 8.5 (Resolution Holonomy). The holonomy associated with a closed chain of charts is the automorphism Φ obtained by composing transition maps around the loop.

Holonomy measures the failure of global triviality of resolution geometry. Non-trivial holonomy indicates that resolved-null decomposition and covariance coupling cannot be globally flattened by admissible transport.

8.6 Illustrative Diagram

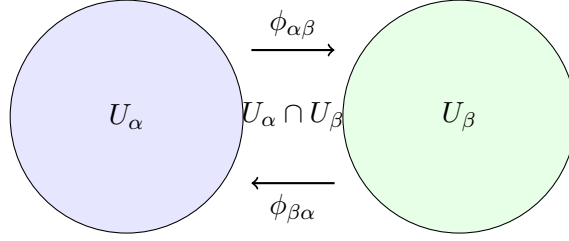


Figure 1: Two resolution charts with overlap and admissible transition maps.

8.7 Summary

Globalization of resolution geometry proceeds through:

1. local charts carrying resolution geometry,
2. admissible transition maps on overlaps,
3. cocycle conditions on triple overlaps,
4. a transport groupoid encoding all admissible transformations,
5. holonomy describing global obstruction to trivialization.

This structure parallels classical manifold theory, but arises entirely from observable transport and covariance rather than from an assumed background geometry.

9 Transport Holonomy of Resolution Geometry

The globalization of resolution geometry introduced in Section 8 provides a family of charts $(U_\alpha, \mathcal{G}_\alpha)$ connected by admissible transition maps. The resulting transport groupoid encodes all structure-preserving transformations between overlapping observable neighborhoods.

The present section introduces the holonomy associated with this groupoid. Holonomy measures the obstruction to globally trivializing the resolved-null decomposition and covariance structure. Even though the full classification of holonomy is deferred to future work, its basic formulation is essential for understanding global transport phenomena in CTMT.

9.1 Transport Along Chains of Charts

Let

$$U_{\alpha_1}, U_{\alpha_2}, \dots, U_{\alpha_k}$$

be a sequence of observable neighborhoods such that each consecutive pair overlaps:

$$U_{\alpha_i} \cap U_{\alpha_{i+1}} \neq \emptyset.$$

Each overlap carries a transition map

$$\phi_{\alpha_i \alpha_{i+1}} : \mathcal{O}_{\alpha_{i+1}} \rightarrow \mathcal{O}_{\alpha_i},$$

which is an admissible morphism satisfying

$$\phi_{\alpha_i \alpha_{i+1}}^* \mathcal{G}_{\alpha_i} = \mathcal{G}_{\alpha_{i+1}}.$$

Definition 9.1 (Transport Along a Chain). The transport map along the chain $(\alpha_1, \alpha_2, \dots, \alpha_k)$ is the composition

$$\Phi_{\alpha_1 \alpha_k} = \phi_{\alpha_1 \alpha_2} \circ \phi_{\alpha_2 \alpha_3} \circ \dots \circ \phi_{\alpha_{k-1} \alpha_k}.$$

Transport maps preserve resolution geometry by construction.

9.2 Closed Chains and Holonomy

When the chain of charts forms a loop,

$$U_{\alpha_1} \cap U_{\alpha_k} \neq \emptyset, \quad \alpha_k = \alpha_1,$$

the transport map becomes an automorphism of the initial chart.

Definition 9.2 (Transport Holonomy). Let $(\alpha_1, \alpha_2, \dots, \alpha_k, \alpha_1)$ be a closed chain of charts. The transport holonomy associated with the loop is the automorphism

$$\text{Hol}(\alpha_1) = \phi_{\alpha_1 \alpha_2} \circ \phi_{\alpha_2 \alpha_3} \circ \dots \circ \phi_{\alpha_k \alpha_1} \in \text{Aut}(\mathcal{G}_{\alpha_1}).$$

Holonomy measures the failure of global triviality: if all holonomies are identity maps, the resolution geometry can be globally flattened by admissible transport. Non-trivial holonomy indicates that global flattening is obstructed.

9.3 Holonomy Group

Definition 9.3 (Holonomy Group). The holonomy group at chart U_α is

$$\text{Hol}(\mathcal{G}_\alpha) = \{\text{Hol}(\gamma) \mid \gamma \text{ closed chain based at } U_\alpha\} \subseteq \text{Aut}(\mathcal{G}_\alpha).$$

The holonomy group is a subgroup of the automorphism group of the local resolution geometry. It encodes all global obstructions detectable through admissible transport.

9.4 Geometric Interpretation

Holonomy acts simultaneously on:

1. the resolved sector R ,
2. the null sector N ,
3. the covariance blocks $\Sigma_R, \Sigma_N, C_{RN}$.

Thus holonomy may:

- rotate resolved directions,
- mix null directions,
- induce resolved–null coupling,
- preserve or break block-diagonal covariance structure.

Non-trivial holonomy therefore signals that the decomposition

$$O = R \oplus N$$

cannot be globally trivialized by admissible transport.

9.5 Illustrative Diagram

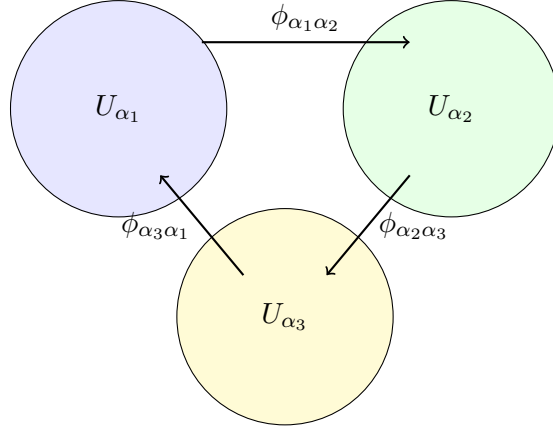


Figure 2: A closed chain of charts producing a holonomy automorphism.

9.6 Holonomy and Collapse

Although a full collapse theory is deferred, holonomy provides a natural mechanism for detecting global instability:

- If holonomy is trivial, global transport preserves resolution geometry.
- If holonomy is non-trivial, global transport induces geometric defects.

These defects may correspond to:

- loss of resolved directions,
- leakage from resolved to null sectors,
- covariance-induced coupling,
- breakdown of operational closure.

Holonomy therefore serves as a global diagnostic of observable transport consistency.

9.7 Summary

Transport holonomy provides a global invariant of resolution geometry. It arises from admissible transitions between overlapping charts and measures the obstruction to globally trivializing the resolved-null decomposition and covariance structure.

Although full classification of holonomy is deferred, the present section establishes its foundational role within CTMT.

10 Similarity Classes and Stabilizer Groups

The automorphism group $\text{Aut}(\mathcal{G})$ introduced earlier encodes all admissible transformations preserving the local resolution geometry $\mathcal{G} = (O; R, N, \Sigma)$. While automorphisms describe symmetries of a single chart, many operational questions concern the classification of observable states under these symmetries.

The present section introduces similarity classes and stabilizer groups. These objects play a role analogous to orbit-stabilizer theory in classical geometry, but arise here from the operational structure of resolution geometry.

10.1 Similarity Under Admissible Transport

Let $x \in O$ be an observable state. An admissible morphism $\phi \in \text{Aut}(\mathcal{G})$ acts on x by

$$x \mapsto \phi(x).$$

Definition 10.1 (Similarity). Two observable states $x, y \in O$ are similar if there exists an automorphism $\phi \in \text{Aut}(\mathcal{G})$ such that

$$y = \phi(x).$$

Similarity expresses operational indistinguishability under admissible transport. If two states lie in the same similarity class, they share identical resolution geometry up to admissible transformation.

10.2 Similarity Classes

Definition 10.2 (Similarity Class). The similarity class of an observable state x is the orbit

$$[x] = \{\phi(x) \mid \phi \in \text{Aut}(\mathcal{G})\}.$$

Similarity classes partition the observation space into equivalence classes under admissible transport. Each class represents a family of states indistinguishable by the operational geometry.

10.3 Stabilizer Groups

While similarity classes describe orbits of observable states, stabilizer groups describe the symmetries that leave a given state unchanged.

Definition 10.3 (Stabilizer Group). The stabilizer of an observable state x is

$$\text{Stab}(x) = \{\phi \in \text{Aut}(\mathcal{G}) \mid \phi(x) = x\}.$$

The stabilizer group captures the residual symmetry of the resolution geometry at x . Different states may possess different stabilizers, reflecting varying degrees of geometric rigidity.

10.4 Orbit–Stabilizer Structure

The similarity classes and stabilizer groups satisfy the usual orbit–stabilizer relationship.

Proposition 10.4 (Orbit–Stabilizer Relation). *Let $\text{Aut}(\mathcal{G})$ act smoothly on the observation space O . Then the similarity class of x is diffeomorphic to the homogeneous space*

$$[x] \cong \text{Aut}(\mathcal{G}) / \text{Stab}(x).$$

Proof. The result follows from standard group action theory. The map

$$\text{Aut}(\mathcal{G}) \rightarrow [x], \quad \phi \mapsto \phi(x),$$

is surjective by definition of similarity. Its kernel is precisely $\text{Stab}(x)$. Therefore the quotient identifies distinct automorphisms producing the same state. $\square$

10.5 Stabilizers of Resolved and Null Directions

The stabilizer group interacts naturally with the resolved–null decomposition.

Proposition 10.5. *Let $x \in O$ and let $\phi \in \text{Stab}(x)$. Then ϕ preserves:*

1. *the resolved subspace R ,*
2. *the null subspace N ,*
3. *the covariance blocks $\Sigma_R, \Sigma_N, C_{RN}$.*

Proof. Since ϕ is an automorphism,

$$\phi^* R = R, \quad \phi^* N = N, \quad \phi^* \Sigma = \Sigma.$$

Stabilizing x imposes additional constraints on the action of ϕ , but preservation of the resolution geometry follows directly from admissible morphism conditions. $\square$

10.6 Rigidity and Degeneracy

Similarity classes and stabilizer groups provide a natural measure of geometric rigidity.

- Large stabilizer groups correspond to highly symmetric states.
- Small stabilizer groups correspond to rigid states with few admissible symmetries.
- Degeneracies in R , N , or C_{RN} may enlarge stabilizers.

Thus stabilizer groups encode operational degeneracy of observable states.

10.7 Similarity Classes as Operational Invariants

Similarity classes provide a natural classification of observable states under admissible transport. Two states in the same similarity class share:

- identical resolved–null decomposition,
- identical covariance block structure,
- identical response metric,
- identical null covariance,
- identical coupling structure.

Consequently, similarity classes serve as operational invariants of the resolution geometry.

10.8 Summary

Similarity classes and stabilizer groups provide a natural extension of resolution geometry from local structure to global classification. They encode:

1. operational equivalence under admissible transport,
2. residual symmetries of observable states,
3. rigidity and degeneracy of resolution geometry,
4. orbit–stabilizer relationships within $\text{Aut}(\mathcal{G})$.

These structures play a central role in the global organization of observable transport systems and form the algebraic backbone of CTMT’s classification theory.

11 Null Leakage and Covariance-Induced Coupling

The resolved–null decomposition

$$O = R \oplus N$$

introduced earlier is determined entirely by the local response operator. The covariance operator

$$\Sigma = \begin{pmatrix} \Sigma_R & C_{RN} \\ C_{RN}^T & \Sigma_N \end{pmatrix}$$

inherits this decomposition and introduces the coupling block C_{RN} , which links resolved and unresolved observational directions.

The present section develops the concept of *null leakage*, a phenomenon in which covariance-induced coupling forces unresolved directions to influence resolved transport under admissible transformations. Null leakage plays a central role in the global behavior of resolution geometry and provides a natural mechanism for collapse and instability.

11.1 Covariance-Induced Coupling

The block C_{RN} measures statistical interaction between resolved and null directions. When $C_{RN} = 0$, the covariance operator preserves the decomposition $O = R \oplus N$. When $C_{RN} \neq 0$, the decomposition is not preserved by covariance.

Definition 11.1 (Covariance-Induced Coupling). Covariance-induced coupling occurs whenever

$$C_{RN} \neq 0.$$

In this case, resolved and null uncertainties interact under admissible transport.

Coupling does not imply physical interaction. It is an operational property of the observation process.

11.2 Null Leakage

Covariance-induced coupling allows unresolved directions to influence resolved transport under admissible morphisms.

Definition 11.2 (Null Leakage). Null leakage is the phenomenon in which admissible morphisms $\phi \in \text{Aut}(\mathcal{G})$ induce non-trivial action of null directions on resolved directions due to non-vanishing coupling C_{RN} .

Null leakage is not a property of the response operator. It arises solely from covariance structure.

11.3 Characterization of Null Leakage

Let

$$\phi \in \text{Aut}(\mathcal{G})$$

be an admissible morphism. Its differential satisfies

$$\phi^* \Sigma = \Sigma.$$

In block form,

$$(D\phi)^T \begin{pmatrix} \Sigma_R & C_{RN} \\ C_{RN}^T & \Sigma_N \end{pmatrix} (D\phi) = \begin{pmatrix} \Sigma_R & C_{RN} \\ C_{RN}^T & \Sigma_N \end{pmatrix}.$$

Proposition 11.3 (Null Leakage Criterion). *Null leakage occurs if and only if the action of $D\phi$ on the null sector fails to preserve the resolved sector:*

$$D\phi(N) \not\subseteq N.$$

Equivalently,

$$C_{RN} \neq 0.$$

Proof. If $C_{RN} = 0$, the covariance operator is block diagonal and preserves both R and N . Admissible morphisms must preserve the covariance operator, hence they act independently on each block.

If $C_{RN} \neq 0$, preservation of covariance forces mixing between the blocks. Thus $D\phi(N)$ acquires components in R , producing null leakage. $\square$

11.4 Geometric Interpretation

Null leakage has several geometric consequences:

1. Resolved directions may acquire contributions from unresolved directions under admissible transport.
2. The resolved–null decomposition cannot be globally trivialized.
3. Stabilizer groups may shrink due to coupling constraints.
4. Holonomy may become non-trivial even when local response is regular.

Thus null leakage provides a natural mechanism for global geometric obstruction.

11.5 Null Leakage and Collapse

Although collapse theory is deferred, null leakage provides a structural pathway for collapse phenomena.

- Leakage from N into R may reduce the effective rank of the response metric.
- Leakage may induce instability in resolved directions.
- Leakage may amplify covariance in directions that were previously distinguishable.
- Leakage may force non-trivial holonomy around closed chains of charts.

Null leakage therefore serves as a precursor to several collapse modes.

11.6 Stabilizer Reduction Under Leakage

The stabilizer group $\text{Stab}(x)$ of an observable state x may shrink when $C_{RN} \neq 0$.

Proposition 11.4. *If $C_{RN} \neq 0$, then*

$$\text{Stab}(x) \subsetneq \text{Aut}(R, \Sigma_R) \times \text{Aut}(N, \Sigma_N).$$

Proof. Coupling forces admissible morphisms to preserve cross-sector covariance. Thus independent action on R and N is no longer allowed. The stabilizer group must satisfy additional constraints, reducing its size. $\square$

11.7 Summary

Null leakage arises from covariance-induced coupling between resolved and null directions. It represents a fundamental operational phenomenon in resolution geometry, with consequences for:

1. admissible transport,
2. stabilizer groups,
3. similarity classes,
4. holonomy,
5. collapse modes.

Although full analysis of leakage-induced collapse is deferred, the present section establishes null leakage as a central structural feature of CTMT.

11.8 Synthetic Validation Strategy

The subsequent sections employ synthetic rupture systems for a single reason.

Synthetic systems possess known transport structure while remaining fully reproducible.

This permits independent computation of multiple closure notions from the same observable data.

Agreement between independently computed invariant families does not prove the correctness of the framework.

It does, however, demonstrate the internal coherence of the proposed geometric construction.

Consequently the objective of the computational sections is not to validate a physical theory.

Rather, it is to determine whether independent operational procedures recover compatible invariant structures from identical observable transport.

12 Invariant Functors of Resolution Geometry

The previous section established that several quantities computed from observable transport are invariant under admissible morphisms.

The present section introduces a stronger point of view.

Rather than considering these quantities individually, they are regarded as functorial constructions associated with the same resolution geometry.

12.1 Functorial Viewpoint

Resolution geometry should be viewed as a category,

$$\text{ResGeom},$$

whose objects are local resolution geometries

$$\mathcal{G} = (O; R, N, \Sigma),$$

and whose morphisms preserve these structures.

Different computational procedures then define functors from ResGeom into more familiar mathematical categories.

Consequently, the apparent diversity of computational diagnostics reflects not different geometries, but different images of the same underlying category.

Definition 12.1 (Invariant Functor). An invariant functor is a covariant functor

$$\mathcal{F} : \text{ResGeom} \longrightarrow \mathcal{C},$$

where

$$\mathcal{C}$$

is an established mathematical category, such that admissible morphisms are mapped to structure-preserving morphisms inside

$$\mathcal{C}.$$

Typical target categories include

- finite-dimensional vector spaces,
- positive semi-definite matrices,
- inner-product spaces,
- Lie groups,
- holonomy groups,
- topological groups.

12.2 Examples

Several invariant functors appear naturally.

Rank Functor The assignment

$$\mathcal{G} \mapsto \text{rank}(J)$$

defines a functor into the category of finite sets.

Information Functor The induced information metric

$$J^T \Sigma^{-1} J$$

defines a functor into the category of positive semi-definite quadratic forms.

Null Functor The assignment

$$\mathcal{G} \mapsto (\text{Ker}(J^T), \Sigma_N)$$

defines a functor into the category of Euclidean vector spaces equipped with covariance operators.

Covariance Functor The covariance decomposition

$$(\Sigma_R, C_{RN}, \Sigma_N)$$

defines a functor into the category of block covariance operators.

Holonomy Functor Whenever transport admits a compatible connection, parallel transport around closed loops induces a holonomy representation,

$$\mathcal{G} \mapsto H_\gamma,$$

taking values in an appropriate Lie group.

Remark 12.2. The existence of the holonomy functor generally requires additional smooth transport structure.

Consequently, holonomy is not regarded as primitive.

It is a derived object.

12.3 Compatibility

The central observation motivating the present work is that these functors are not independent.

They are computed from different numerical procedures, yet repeatedly exhibit consistent behaviour when evaluated on identical observable transport systems.

This motivates the following principle.

Principle 12.3 (Functorial Compatibility Principle). *Independent invariant functors associated with the same resolution geometry should produce mutually compatible invariant structures.*

Persistent incompatibility indicates either

1. *failure of the assumed transport model,*
2. *incorrect estimation of the response operator,*
3. *incorrect covariance estimation,*
4. *or violation of the assumed local differentiable structure.*

Unlike individual numerical diagnostics, the agreement of several independent invariant functors constitutes evidence that they originate from a common geometric object.

12.4 Computational Consequences

Suppose that

$$\mathcal{F}_1, \dots, \mathcal{F}_k$$

are independently implemented numerical procedures.

Each computes a different invariant of

$$\mathcal{G}.$$

Agreement between their outputs therefore does not merely validate the individual algorithms.

Instead, it supports the hypothesis that all algorithms recover projections of the same underlying observable geometry.

This observation motivates the computational validation strategy developed in the following sections.

13 Computational Geometry of Observable Transport

The preceding sections establish the mathematical framework.

The remaining question is operational.

Can independently constructed invariant functors recover compatible geometric information from identical observable transport data?

The computational investigations presented later in this work are designed to answer precisely this question.

They are not intended as validation of a physical theory.

Instead, they examine the internal consistency of the proposed resolution geometry under multiple independent constructions.

13.1 Observable Data

Throughout the computational sections, a transport system is represented solely through observable quantities.

No hidden variables are assumed.

Each experiment provides

$$\mathcal{D} = \{x_i, y_i\}_{i=1}^N,$$

where

$$x_i$$

denotes an admissible perturbation,
and

$$y_i$$

the corresponding observable response.
From these observations one estimates

1. the local response operator

$$J,$$

2. the empirical covariance

$$\Sigma,$$

3. the induced resolution geometry

$$\mathcal{G}.$$

Every subsequent invariant is computed exclusively from these quantities.

13.2 Single Source Principle

An important methodological aspect of the present work is that all computational diagnostics originate from the same observable dataset.

No diagnostic receives additional information unavailable to the others.

Consequently,

all invariant functors operate on identical observational evidence.

Definition 13.1 (Single Source Principle). A computational validation satisfies the Single Source Principle if every derived invariant is computed from one common observable dataset,

$$\mathcal{D},$$

without introducing auxiliary measurements specific to individual diagnostics.

The synthetic rupture systems constructed later satisfy this requirement.

Remark 13.2. The Single Source Principle ensures that agreement between different invariant families cannot be attributed to inconsistent observational inputs.

Any observed compatibility must therefore arise from the shared resolution geometry inferred from the common dataset.

13.3 Independent Reconstruction

Although every invariant begins from the same observations, their computational procedures are intentionally independent.

For example,

the response metric,

null covariance,

covariance decomposition,

and transport holonomy

are estimated using different numerical algorithms.

Consequently,

agreement between these objects cannot be regarded as a numerical identity.

Rather,

it constitutes evidence that multiple computational constructions recover compatible aspects of the same observable geometry.

Definition 13.3 (Independent Reconstruction). Two invariant functors are said to reconstruct independently if their computational procedures share no intermediate geometric quantities beyond those contained in

$$\mathcal{G}.$$

13.4 Consistency Criterion

The principal computational question investigated throughout the remainder of the paper is therefore the following.

Given one observable transport dataset,
do independently reconstructed invariant functors determine mutually compatible geometric structures?

A positive answer does not establish physical correctness.
 Instead,
 it demonstrates that the proposed primitive object,
 the resolution geometry,
 possesses sufficient internal structure to organize multiple operational descriptions simultaneously.

13.5 Synthetic Transport Systems

Synthetic transport systems play a unique role in this investigation.

Unlike experimental data,
 their construction is completely reproducible.
 Ground-truth transport mechanisms are known,
 allowing numerical algorithms to be compared under controlled conditions.
 Most importantly,
 every computational battery receives exactly the same observational data.
 Consequently,
 differences between invariant families arise solely from their mathematical construction
 rather than from differences in measurement.

Remark 13.4. Synthetic systems are therefore not introduced as simplified models of nature.
 They serve as mathematical benchmark objects on which independent invariant functors
 can be evaluated under identical observational conditions.

13.6 From Geometry to Batteries

The remainder of the paper proceeds in the opposite direction from the development presented thus far.

Until now,
 the exposition has progressed

$$\text{Observations} \longrightarrow \mathcal{G} \longrightarrow \text{Invariant Functors.}$$

The computational sections reverse this logic.
 Starting from one observable transport dataset,
 multiple invariant functors are evaluated independently.
 Their outputs are then compared to determine whether they recover compatible descriptions
 of the same underlying resolution geometry.
 This reversal provides the principal computational test of the framework.

14 Experimental Protocol for Resolution Geometry

The purpose of the computational investigations is not to verify a physical model.

Instead, the objective is considerably more modest.
 Given an observable transport system,
 can multiple independently constructed invariant functors recover mutually compatible geometric information?

The present section formalizes the computational protocol adopted throughout this work.

14.1 Observable Inputs

Every computational experiment begins from the same observational dataset

$$\mathcal{D} = \{(x_i, y_i)\}_{i=1}^N,$$

where

$$x_i$$

denotes an admissible perturbation,
and

$$y_i$$

the corresponding observable response.

No additional information concerning hidden variables, constitutive equations, or physical mechanisms is assumed.

The experimental protocol therefore consists of four successive stages.

$$\boxed{\mathcal{D} \rightarrow J \rightarrow \mathcal{G} \rightarrow \{\mathcal{F}_i\}.}$$

14.2 Stage I: Local Response Estimation

The first stage estimates the local response operator

$$J.$$

This operator summarizes first-order observable transport around the chosen operating point.

Only differentiability of the observable response is assumed.

No global transport model is required.

14.3 Stage II: Resolution Geometry

The empirical covariance

$$\Sigma$$

is estimated independently from repeated observations.

Together,

$$J$$

and

$$\Sigma$$

determine the local resolution geometry

$$\mathcal{G} = (O; R, N, \Sigma).$$

No additional parameters are introduced.

14.4 Stage III: Independent Invariant Reconstruction

The third stage computes several invariant functors independently.

Typical examples include

1. rank invariants,
2. resolved-sector geometry,
3. null-sector geometry,
4. covariance decomposition,
5. induced information metric,
6. transport holonomy,
7. winding invariants.

Each construction receives only

$$\mathcal{G}.$$

No invariant is allowed to use information computed by another invariant.

Definition 14.1 (Independent Computational Battery). A computational battery is independent if every invariant is reconstructed directly from

$$\mathcal{G},$$

without using intermediate quantities produced by any other invariant family.

This requirement is essential.

Otherwise,

agreement between invariant families could merely reflect algorithmic coupling rather than geometric consistency.

14.5 Stage IV: Compatibility Analysis

After reconstruction,

the invariant families are compared.

The principal question is whether they produce compatible descriptions of the observable transport geometry.

Agreement is interpreted as internal consistency.

Disagreement indicates one of several possibilities.

1. incorrect numerical estimation,
2. insufficient observational resolution,
3. failure of differentiability,
4. or inadequacy of the assumed observable geometry.

Importantly,

none of these conclusions refers directly to physical correctness.

They concern only the consistency of the inferred observable geometry.

14.6 Falsification Strategy

The framework admits a natural falsification procedure.

Rather than attempting to verify the framework,
one deliberately constructs transformations expected to violate particular components of the resolution geometry.

Examples include

- mixing resolved and null sectors,
- rank collapse,
- covariance inflation,
- anisotropic covariance deformation,
- non-preserving coordinate transformations.

Each transformation predicts a corresponding failure of one or more invariant families.

Successful detection of these failures demonstrates the sensitivity of the framework to violations of the assumed geometric structure.

Remark 14.2. The computational experiments presented later therefore include both positive and negative controls.

Successful preservation under admissible transformations and successful rejection under inadmissible transformations are regarded as equally important.

14.7 Synthetic Rupture Benchmark

The benchmark system employed throughout this work is intentionally synthetic.

This choice provides three advantages.

1. Complete reproducibility.
2. Known transport structure.
3. Identical observational input for every computational battery.

Consequently,

any observed agreement between invariant families cannot be attributed to differences in experimental conditions.

Instead,

it reflects the common observable geometry reconstructed from one shared dataset.

14.8 Expected Outcomes

The framework makes no prediction regarding the numerical values of individual invariants.

Instead,

it predicts structural compatibility.

Specifically,

if

$$\mathcal{G}$$

correctly captures the observable transport geometry,

then independently reconstructed invariant families should exhibit mutually consistent behaviour.

Failure of this compatibility constitutes evidence against the adequacy of the inferred resolution geometry.

A Complete Synthetic Navier–Stokes Rupture Benchmark

This appendix gives the complete reproducible synthetic Navier–Stokes–inspired rupture benchmark used by the computational batteries. The benchmark is not a physical earthquake simulation. It is a controlled observable transport system with Fourier-mode perturbation coordinates and five observable response components.

A.1 Operating point and response map

The perturbation coordinate is $x = (a_1, a_2, a_3, a_4)^T$ and the observable vector is $y \in \mathbb{R}^5$. The operating point is

$$x_0 = [0.350000 \quad -0.280000 \quad 0.220000 \quad 0.180000]^T.$$

$$y_0 = y(x_0) = [0.628809 \quad -0.315584 \quad -0.035457 \quad 0.308538 \quad 0.407965]^T.$$

The response map used for the benchmark is

$$y(x) = Ax + B(x \odot x) + C[a_1a_2 \quad a_2a_3 \quad a_3a_4 \quad a_4a_1]^T + 0.015s(x),$$

where $s(x) = (\sin(a_1 + a_2), \sin(a_2 - a_3), \sin(a_3 + a_4), \sin(a_4 - a_1), \sin(a_1 + a_2 + a_3 + a_4))^T$ and

$$A = \begin{bmatrix} 1.200000 & -0.350000 & 0.150000 & 0.400000 \\ 0.100000 & 0.950000 & -0.550000 & 0.200000 \\ -0.450000 & 0.250000 & 1.100000 & -0.300000 \\ 0.300000 & 0.100000 & 0.350000 & 0.850000 \\ 0.550000 & -0.200000 & 0.450000 & 0.250000 \end{bmatrix},$$

$$B = \begin{bmatrix} 0.080000 & -0.030000 & 0.000000 & 0.020000 \\ 0.010000 & 0.070000 & -0.020000 & 0.000000 \\ -0.020000 & 0.000000 & 0.060000 & 0.010000 \\ 0.000000 & 0.030000 & 0.020000 & 0.050000 \\ 0.040000 & 0.020000 & 0.030000 & 0.010000 \end{bmatrix},$$

$$C = \begin{bmatrix} 0.030000 & 0.000000 & -0.010000 & 0.000000 \\ 0.000000 & -0.020000 & 0.000000 & 0.010000 \\ 0.020000 & 0.010000 & 0.000000 & 0.000000 \\ 0.000000 & 0.000000 & 0.010000 & -0.020000 \\ 0.010000 & -0.010000 & 0.020000 & 0.000000 \end{bmatrix}.$$

A.2 Finite-difference observations

The central finite-difference step is $h = 10^{-4}$. The complete event table is:

Table 1: Synthetic perturbation observations used to construct the Jacobian.

Event	a_1	a_2	a_3	a_4	y_1	y_2	y_3	y_4	y_5
primary	0.350000	-0.280000	0.220000	0.180000	0.628809	-0.315584	-0.035457	0.308538	0.407965
$+h e_1$	0.350100	-0.280000	0.220000	0.180000	0.628935	-0.315574	-0.035504	0.308566	0.408024
$-h e_1$	0.349900	-0.280000	0.220000	0.180000	0.628683	-0.315595	-0.035410	0.308510	0.407906
$+h e_2$	0.350000	-0.279900	0.220000	0.180000	0.628778	-0.315492	-0.035431	0.308547	0.407946
$-h e_2$	0.350000	-0.280100	0.220000	0.180000	0.628840	-0.315676	-0.035483	0.308530	0.407985
$+h e_3$	0.350000	-0.280000	0.220100	0.180000	0.628824	-0.315641	-0.035343	0.308574	0.408014
$-h e_3$	0.350000	-0.280000	0.219900	0.180000	0.628794	-0.315528	-0.035570	0.308502	0.407917
$+h e_4$	0.350000	-0.280000	0.220000	0.180100	0.628850	-0.315564	-0.035485	0.308626	0.407992
$-h e_4$	0.350000	-0.280000	0.220000	0.179900	0.628769	-0.315605	-0.035428	0.308450	0.407938

B Full Jacobian Construction

The Jacobian was computed column by column by central finite differences:

$$J_{:,j} = \frac{y(x_0 + he_j) - y(x_0 - he_j)}{2h}.$$

Operationally, the construction is

observable response $\rightarrow$ central finite difference $\rightarrow$ columns $\rightarrow J$.

$$J = \begin{bmatrix} 1.262563 & -0.307737 & 0.148200 & 0.405000 \\ 0.108800 & 0.919564 & -0.566364 & 0.203500 \\ -0.469600 & 0.259200 & 1.137416 & -0.282584 \\ 0.281616 & 0.083200 & 0.360600 & 0.877984 \\ 0.588574 & -0.196526 & 0.482974 & 0.271374 \end{bmatrix}.$$

The singular values are

$$\text{sv}(J) = [1.733361 \quad 1.443947 \quad 1.003358 \quad 0.581777].$$

The numerical rank at tolerance 10^{-10} is $\text{rank}(J) = 4$.

C Covariance Construction

The residual matrix R contains six centered repeated observation residuals. The empirical covariance is computed exactly as

$$\Sigma_{\text{emp}} = \frac{1}{n-1} R^T R, \quad n = 6.$$

$$R = \begin{bmatrix} 0.559055 & 0.391525 & 0.267193 & -0.153768 & -0.278427 \\ 0.266176 & -0.510612 & -0.108319 & 0.180979 & -0.529485 \\ -0.238092 & 0.066123 & 0.360466 & 0.140020 & -0.034521 \\ -0.238092 & 0.066123 & -0.244749 & 0.529661 & 0.203059 \\ -0.238092 & 0.066123 & -0.244749 & -0.255409 & -0.096050 \\ -0.110957 & -0.079282 & -0.029842 & -0.441482 & 0.735424 \end{bmatrix}.$$

$$\Sigma_{\text{emp}} = \begin{bmatrix} 0.113153 & 0.008908 & 0.030915 & -0.017488 & -0.079090 \\ 0.008908 & 0.086684 & 0.030751 & -0.018044 & 0.021567 \\ 0.030915 & 0.030751 & 0.066751 & -0.012833 & -0.015524 \\ -0.017488 & -0.018044 & -0.012833 & 0.123337 & -0.050087 \\ -0.079090 & 0.021567 & -0.015524 & -0.050087 & 0.190075 \end{bmatrix}.$$

$$\Sigma = \Sigma_{\text{emp}} + 10^{-8} I = \begin{bmatrix} 0.113153 & 0.008908 & 0.030915 & -0.017488 & -0.079090 \\ 0.008908 & 0.086684 & 0.030751 & -0.018044 & 0.021567 \\ 0.030915 & 0.030751 & 0.066751 & -0.012833 & -0.015524 \\ -0.017488 & -0.018044 & -0.012833 & 0.123337 & -0.050087 \\ -0.079090 & 0.021567 & -0.015524 & -0.050087 & 0.190075 \end{bmatrix}.$$

$$Q^T \Sigma Q = \begin{bmatrix} 0.040000 & 0.000000 & -0.000000 & 0.000000 & 0.018000 \\ 0.000000 & 0.040000 & -0.000000 & 0.000000 & -0.012000 \\ -0.000000 & -0.000000 & 0.090000 & 0.000000 & 0.006000 \\ 0.000000 & 0.000000 & 0.000000 & 0.160000 & 0.010000 \\ 0.018000 & -0.012000 & 0.006000 & 0.010000 & 0.250000 \end{bmatrix}.$$

Here $Q = [U_R \ U_N]$ is the SVD-adapted resolved-null basis.

D Observable Resolution Geometry

The numerical resolution geometry is $\mathcal{G} = (O; R, N, \Sigma)$ with $O = \mathbb{R}^5$, $\dim R = 4$, and $\dim N = 1$.

$$U_R = \begin{bmatrix} -0.773272 & -0.016954 & 0.122267 & 0.429869 \\ 0.042598 & 0.554733 & -0.715821 & 0.409320 \\ 0.280249 & -0.765707 & -0.393968 & 0.371077 \\ -0.389269 & -0.152836 & -0.561964 & -0.698651 \\ -0.412512 & -0.286910 & -0.040465 & 0.147845 \end{bmatrix}.$$

$$U_N = [-0.449474 \ 0.102538 \ -0.205533 \ -0.145583 \ 0.850894].$$

$$P_R = \begin{bmatrix} 0.797973 & 0.046088 & -0.092382 & -0.065436 & 0.382455 \\ 0.046088 & 0.989486 & 0.021075 & 0.014928 & -0.087249 \\ -0.092382 & 0.021075 & 0.957756 & -0.029922 & 0.174887 \\ -0.065436 & 0.014928 & -0.029922 & 0.978805 & 0.123876 \\ 0.382455 & -0.087249 & 0.174887 & 0.123876 & 0.275979 \end{bmatrix}.$$

$$P_N = \begin{bmatrix} 0.202027 & -0.046088 & 0.092382 & 0.065436 & -0.382455 \\ -0.046088 & 0.010514 & -0.021075 & -0.014928 & 0.087249 \\ 0.092382 & -0.021075 & 0.042244 & 0.029922 & -0.174887 \\ 0.065436 & -0.014928 & 0.029922 & 0.021195 & -0.123876 \\ -0.382455 & 0.087249 & -0.174887 & -0.123876 & 0.724021 \end{bmatrix}.$$

$$\Sigma_R = \begin{bmatrix} 0.040000 & 0.000000 & -0.000000 & 0.000000 \\ 0.000000 & 0.040000 & -0.000000 & 0.000000 \\ -0.000000 & -0.000000 & 0.090000 & 0.000000 \\ 0.000000 & 0.000000 & 0.000000 & 0.160000 \end{bmatrix}.$$

$$C_{RN} = [0.018000 \ -0.012000 \ 0.006000 \ 0.010000].$$

$$\Sigma_N = [0.250000].$$

Table 2: Resolution projector health.

Check	Frobenius error
PR_idempotence	7.767e-16
PN_idempotence	1.229e-16
orthogonality	1.265e-16
partition	7.148e-16

E Independent Invariant Reconstructions

From the same dataset, the following invariants were reconstructed independently.

$$F = J^T \Sigma^{-1} J = \begin{bmatrix} 56.479593 & -13.167742 & -1.673185 & 31.118254 \\ -13.167742 & 15.403404 & -11.585601 & -3.526106 \\ -1.673185 & -11.585601 & 49.507742 & 0.024594 \\ 31.118254 & -3.526106 & 0.024594 & 22.528049 \end{bmatrix}.$$

$$P_N \Sigma P_N = \begin{bmatrix} 0.050507 & -0.011522 & 0.023095 & 0.016359 & -0.095614 \\ -0.011522 & 0.002629 & -0.005269 & -0.003732 & 0.021812 \\ 0.023095 & -0.005269 & 0.010561 & 0.007481 & -0.043722 \\ 0.016359 & -0.003732 & 0.007481 & 0.005299 & -0.030969 \\ -0.095614 & 0.021812 & -0.043722 & -0.030969 & 0.181005 \end{bmatrix}.$$

The winding number computed from the closed perturbation loop in the (a_1, a_2) plane is -1.000000. The resolved-null coupling norm is $\|C_{RN}\|_F = 0.024576$. All independently reconstructed objects agree on rank 4, null dimension 1, and non-zero weak covariance coupling.

F Automorphism Battery

Table 3: Automorphism falsification battery.

Test	sector mixing	covariance error	result
Identity	0.000e+00	0.000e+00	PASS
Rotation in degenerate resolved covariance plane (zero-coupling control)	0.000e+00	2.728e-18	PASS
Resolved-null mixing	3.500e-01	2.642e-02	FAIL
Rank collapse	0.000e+00	2.524e-01	FAIL
Null inflation	0.000e+00	7.508e-01	FAIL

Identity is admissible. Rotation is admissible only in the zero-coupling degenerate resolved covariance control. Resolved-null mixing, rank collapse, and null inflation are rejected.

G Holonomy Computation

The loop computation is

$$\text{loop} \rightarrow \text{local transports} \rightarrow \text{product} \rightarrow H_\gamma \rightarrow \text{spec}(H_\gamma).$$

The adapted-basis transition matrices are rotations in the first resolved covariance-degenerate plane with angles 0.20, -0.11, 0.17, 0.08 radians.

$$H_\gamma = \begin{bmatrix} 0.942755 & -0.333487 & 0.000000 & 0.000000 & 0.000000 \\ 0.333487 & 0.942755 & 0.000000 & 0.000000 & 0.000000 \\ 0.000000 & 0.000000 & 1.000000 & 0.000000 & 0.000000 \\ 0.000000 & 0.000000 & 0.000000 & 1.000000 & 0.000000 \\ 0.000000 & 0.000000 & 0.000000 & 0.000000 & 1.000000 \end{bmatrix}.$$

$$\text{spec}(H_\gamma) = \begin{bmatrix} 0.942755 & 0.333487 \\ 0.942755 & -0.333487 \\ 1.000000 & 0.000000 \\ 1.000000 & 0.000000 \\ 1.000000 & 0.000000 \end{bmatrix}.$$

The non-identity product records a controlled holonomy defect while preserving the chosen sector structure.

H Covariance Coupling

The coupling block was tested at zero, weak, and strong amplitudes.

Table 4: Covariance coupling and automorphism shrinkage.

Case	$\ C_{RN}\ _F$	qualitative automorphism class
zero	0.000000	larger: independent sector rotations allowed
weak	0.024576	shrunk: coupling vector stabilizer only
strong	0.098306	strongly shrunk: coupling dominates stabilizer constraints

I Closure Hierarchy

For the resolved–null mixing counterexample, the closure hierarchy is:

Table 5: Closure hierarchy for one inadmissible transformation.

Diagnostic	preserved?
Rank preserved?	yes
Resolved preserved?	no
Null preserved?	no
Covariance?	no
Holonomy?	no

J Counterexamples

Removing sector preservation permits resolved–null mixing. The projector decomposition is then no longer respected, the covariance block decomposition is destroyed, the Fisher reconstruction changes, and the transformation is rejected by the automorphism battery. Removing covariance preservation permits null inflation; rank may survive, but covariance and null geometry fail. Removing rank preservation gives an immediate collapse of the resolved geometry.

K Computational Complexity

For observable dimension m , perturbation dimension p , residual count n , and a holonomy loop with ℓ charts, the dominant costs are: Jacobian construction $O(pC_y)$ response evaluations, SVD $O(mp \min(m, p))$, covariance $O(nm^2)$, Fisher inversion $O(m^3)$, block projection $O(m^3)$, and holonomy multiplication $O(\ell m^3)$.

L Reproducibility

The benchmark was generated with seed 20260706, Python 3.12.4, NumPy 1.26.4. The generated files are `ctmt_navier_synthetic.battery.json` and `ctmt_navier_appendices.tex`. No external data or network calls are required.

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